

CommonJS vs ES Modules in Node.js

Node.js supports two module systems. CommonJS (CJS) is the original system and uses **require()** and **module.exports**. ES Modules (ESM) are the JavaScript standard and use **import** and **export**.

How Node.js decides which system a file uses

Files ending in **.mjs** are always ESM and **.cjs** are always CommonJS. For **.js** files, Node.js reads the nearest package.json: **"type": "module"** means ESM, otherwise CommonJS.

Key differences

Feature	CommonJS	ES Modules
Syntax	require / module.exports	import / export
Loading	synchronous	asynchronous
Top-level await	not supported	supported
__dirname	available	use import.meta.dirname
File extension in imports	optional	required ('./util.js')

Loading ESM from CommonJS

Older Node.js versions throw **ERR_REQUIRE_ESM** when require() loads an ES module. Use a dynamic import instead, which works in both systems:

```
const { default: chalk } = await import('chalk');
```